

Hospital Chatbot

for Color Coded Clinical Pathways

Presenter Walkthrough & Speaker Script

Step-by-step guide for all 20 slides in `main.pdf`

Quaigraine Samuel
Twum Samuel

Department of Mathematics, KNUST

August 2026

Project **main.pdf** on the projector; keep **this handbook** on your laptop or print it. Each section matches one Beamer slide in order. Read the **SAY THIS** boxes aloud or paraphrase.

Contents

1	How to use this handbook	2
2	Slide-by-slide walkthrough	3
2.1	Slide 1 — Title	3
2.2	Slide 2 — Outline of Presentation	3
2.3	Slide 3 — ED overcrowding and waiting for care	3
2.4	Slide 4 — Research gaps in prior work	4
2.5	Slide 5 — Case study setting	5
2.6	Slide 6 — Dataset schema	5
2.7	Slide 7 — Acuity distribution	6
2.8	Slide 8 — Pathway destinations	6
2.9	Slide 9 — Green-pathway diversion	7
2.10	Slide 10 — Chatbot colour-routing accuracy	8
2.11	Slide 11 — Case study conclusion	9
2.12	Slide 12 — ED overcrowding as a network problem	9
2.13	Slide 13 — Hospital transshipment network	10
2.14	Slide 14 — LP structure	10
2.15	Slide 15 — Acute-zone load reduction	11
2.16	Slide 16 — Conversational NLP supports triage	11
2.17	Slide 17 — Versus generative clinical chatbots	12
2.18	Slide 18 — Summary	12
2.19	Slide 19 — Limitations	13
2.20	Slide 20 — References	13
3	Anticipated Q&A	14
4	Rebuild commands	14

1 How to use this handbook

- **Structure.** Twenty slides in `community-hospital/slides/main.pdf`, in fixed order. This document has one subsection per slide.
- **Coloured boxes.**
 - **ON SCREEN** — what the audience sees; point at it while speaking.
 - **THE IDEA** — plain-language concept (math/OR slides).
 - **SAY THIS** — full spoken script with verbal citations.
- **Citations.** Slides show author–year in parentheses; name them aloud (e.g. “Mitchell and colleagues, twenty twenty-three”).
- **Our numbers.** Say “in our Community Hospital case study” for metrics tagged This study, 2026.

Pre-flight checklist

1. Open `main.pdf` full-screen on the projector.
2. Have this PDF on a second screen or printed.
3. Confirm chart images render (slides 7–10, 15).
4. Optional: open `community_hospital_patients.csv` if asked about data.

Timing overview

Slides	Section	Slides	Minutes
1–2	Opening	2	2
3–4	Research gaps	2	4
5–11	Community Hospital case study	7	10
12–15	Transshipment optimization	4	6
16–17	NLP triage design	2	4
18–20	Conclusion & references	3	4
Total		20	25–30

Key numbers cheat sheet

- $N = 1000$; skewed mix 5% / 15% / 30% / 25% / 25%
- Colour-routing accuracy: **97.5%** (975/1000)
- Level 1 recall: **94.0%** (47/50)
- Green diversion: **500** patients (50%)
- Transshipment: cost 7150 $\rightarrow$ 2700 (**62%**); acute load 900 $\rightarrow$ 500 (**44%**)
- 8k BioBERT holdout reference: 99.92%

2 Slide-by-slide walkthrough

2.1 Slide 1 — Title

main.pdf slide: 1 -- Opening

Suggested: 30 sec

ON SCREEN

Title: *Hospital Chatbot for Color Coded Clinical Pathways*. Subtitle: Community Hospital case study. Authors: Quaigraine Samuel, Twum Samuel. KNUST Mathematics, August 2026.

SAY THIS

Good morning everyone. Thank you for being here. Today we present our final-year project: **Hospital Chatbot for Color Coded Clinical Pathways**. We use a synthetic Community Hospital case study in Ghana to show how free-text chief complaints can map to SATS colours and pathway cards, and how that can reduce emergency-department overcrowding. I am Quaigraine Samuel, and my colleague is Twum Samuel, from the Department of Mathematics at KNUST.

Transition: Advance to the outline.

2.2 Slide 2 — Outline of Presentation

main.pdf slide: 2 -- Opening

Suggested: 1 min

ON SCREEN

Table of contents: Research gaps; Community Hospital case study; Transshipment optimization; NLP triage design; Conclusion; References.

SAY THIS

Here is the road map. We start with **research gaps** in prior work—why text-based pre-triage matters for overcrowding. Then our **Community Hospital** synthetic evaluation with a thousand skewed cases. Next, **transshipment optimization** models patient flow as a network problem. We explain **NLP design**—conversational intake but constrained outputs, not a generic chatbot. We close with **summary, limitations**, and full **references**.

Transition: Open with real-world ED overcrowding images.

2.3 Slide 3 — ED overcrowding and waiting for care

main.pdf slide: 3 -- Research gaps

Suggested: 2 min

ON SCREEN

Two photos side by side with source lines:

- Left: Longdon (2026), Citi Newsroom — queue for care.
- Right: Aminu (2021), Ghana Fact — Korle-Bu overcrowding fact-check.

**SAY THIS**

Before the literature, look at the reality. On the left, Bervelyn Longdon's twenty twenty-six Citi Newsroom piece on *queue for care*—patients waiting because the system is stretched. On the right, Ghana Fact's twenty twenty-one fact-check on Korle-Bu Teaching Hospital overcrowding: hallways and wards under pressure. These are not abstract statistics; they are why we care about **routing the right patient to the right queue early**. Our project asks: can we estimate urgency from words *before* vitals and the physical triage desk, to reduce this congestion?

Transition: State the five literature gaps on one slide.

2.4 Slide 4 — Research gaps in prior work

main.pdf slide: 4 -- Research gaps

Suggested: 2–3 min

ON SCREEN

Intro sentence plus five numbered one-line gaps, each with author–year citations (SATS 2012; Rominski et al. 2014; Stewart et al. 2023; Mitchell et al. 2023; Yilmaz 2021; etc.).

SAY THIS

Our aim is to infer acuity from free-form chief complaints to reduce ED overcrowding and waiting times— Mitchell and colleagues in twenty twenty-three and Yilmaz in twenty twenty-one frame that problem.

Gap one, vitals-first bias: SATS and TEWS assume bedside physiology first; Stewart's twenty twenty-three NLP review shows few systems estimate acuity from text when vitals are still unknown.

Gap two, pathway discontinuity: colours are assigned at the desk, but Nurchis et al. twenty twenty-five and Sundström et al. twenty fourteen show waiting-time and destination protocols are not always carried forward digitally.

Gap three, overcrowding without text pre-triage: Green-tag patients congest acute zones— Yilmaz and Mitchell again.

Gap four, generative versus constrained NLP: open-ended chatbots raise hallucination risk; Stewart and Cheng et al. twenty twenty-five contrast that with fixed acuity classes.

Gap five, class imbalance: papers report overall accuracy but under-report recall on rare Red cases—Stewart and Berkowitz et al. on under-triage safety.

These are gaps in *prior work*; our case study addresses them next.

Transition: Introduce the Community Hospital synthetic cohort.

2.5 Slide 5 — Case study setting

main.pdf slide: 5 -- Community Hospital

Suggested: 2 min

ON SCREEN

Bullets: fictional Kumasi-region LMIC hospital; $N = 1000$ skewed mix; Ghanaian complaints; SATS labels and pathways; 7% ambiguity flag; 97.5% colour accuracy; 99.92% holdout reference.

SAY THIS

We built a **Community Hospital**—a fictional but Kumasi-region-style LMIC facility, grounded in Rominski’s KATH SATS work and Gyedu on trainee triage in Ghana. We generated **one thousand** synthetic patients with a **realistic skewed mix**: five, fifteen, thirty, twenty-five, twenty-five percent across acuity levels—not a balanced toy dataset, following Mitchell on LMIC ED statistics.

Each row has Ghanaian patient-voice complaints—Kejetia, Bantama, Ejisu, Sunyani, KATH references. Nurse ground truth maps level to SATS colour and pathway card. About **seven percent** are flagged borderline. In our case study the chatbot achieves **ninety-seven point five percent** colour-routing accuracy; separately, BioBERT on an eight-thousand holdout reaches ninety-nine point ninety-two percent—Lee et al. twenty twenty on BioBERT.

Transition: Show the CSV schema briefly.

2.6 Slide 6 — Dataset schema

main.pdf slide: 6 -- Community Hospital

Suggested: 1 min

ON SCREEN

Table: patient_id, demographics, chief_complaint, acuity_level, sats_colour, pathway_destination, t_max_minutes, ambiguity_flag. Path to CSV file.

SAY THIS

This slide is for transparency. Every synthetic patient row carries an ID, demographics, the full chief complaint text, nurse acuity level one to five, SATS colour, **pathway destination**, **time target** in minutes, and an ambiguity flag. The file lives at `community-hospital/data/community_hospital_patients.csv`. The point is: we do not only predict a colour—each row binds colour to operational routing data.

Transition: Show the acuity bar chart.

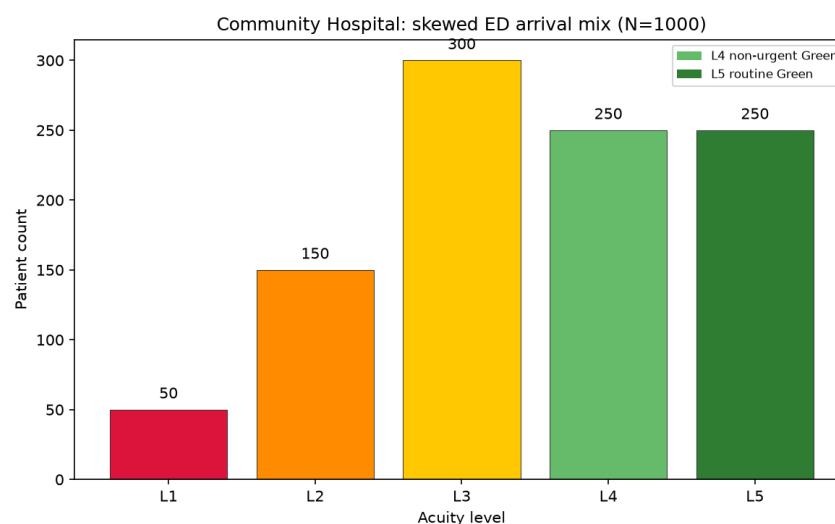
2.7 Slide 7 — Acuity distribution

main.pdf slide: 7 -- Community Hospital

Suggested: 1–2 min

ON SCREEN

Bar chart `acuity_distribution.png`. Caption: skewed mix; L4/L5 distinct shades, both Green.



SAY THIS

This chart shows our **skewed arrival mix**—few critical Reds, many Yellows and Greens, as in real EDs per Mitchell and Rominski. Notice levels four and five use **different bar colours** but both map to SATS Green—that matters when we discuss accuracy by level later.

Transition: Show pathway flow chart.

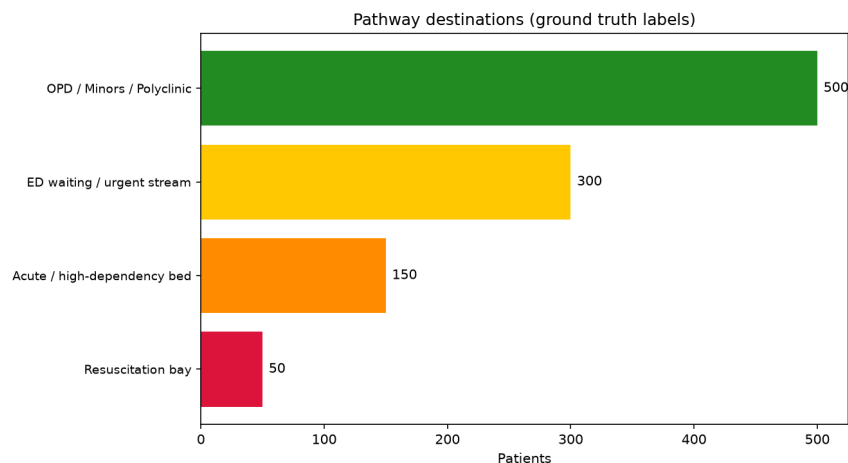
2.8 Slide 8 — Pathway destinations

main.pdf slide: 8 -- Community Hospital

Suggested: 1–2 min

ON SCREEN

Chart `pathway_flow.png`. Plain block: every row has destination and $T_{\max}$; colour binds to protocol.

**SAY THIS**

Colour is not a sticker—it must mean a **protocol**. Sundström et al. on pathway care and the SATS manual show destination and time target matter. Here, Orange routes to acute beds within ten minutes; Green to OPD or minors within hours. Every CSV row carries that pathway card, not just a label.

Transition: Discuss Green diversion and overcrowding.

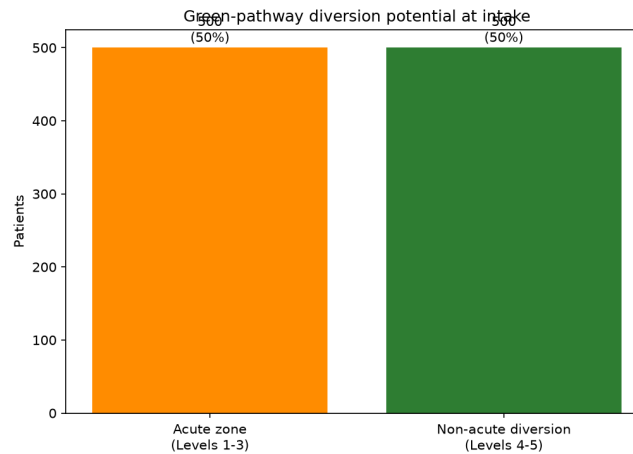
2.9 Slide 9 — Green-pathway diversion

main.pdf slide: 9 -- Community Hospital

Suggested: 2 min

ON SCREEN

Left: `green_diversion.png`. Right: 50% of cohort ($n = 500$) to OPD/Minors; chatbot assigns Green at intake.



SAY THIS

Half our cohort—five hundred patients at levels four and five—are **Green diversion candidates** routed to OPD, Minors, or Polyclinic in our case study. Yilmaz’s Green-tag burden and Mitchell’s overcrowding work explain why that matters: without early diversion, non-urgent patients queue in acute zones. Our chatbot assigns Green **at intake**, before the physical desk—Sundström-style pathway routing applied early.

Transition: Show per-level accuracy chart.

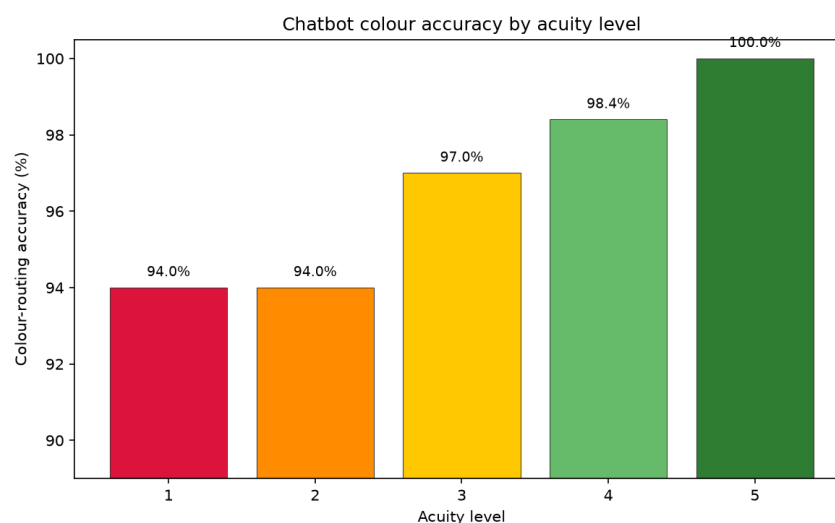
2.10 Slide 10 — Chatbot colour-routing accuracy

main.pdf slide: 10 -- Community Hospital

Suggested: 2 min

ON SCREEN

Chart chatbot_accuracy_by_level.png. 97.5% overall; L1 recall 94% (47/50).



SAY THIS

Overall **ninety-seven point five percent** colour-routing—nine hundred seventy-five of one thousand. Level-one recall is **ninety-four percent**—forty-seven of fifty Reds—the safety-critical minority under skewed priors, as Stewart and Berkowitz warn. Errors cluster on **adjacent levels** and the seven percent borderline flag rows. We report per-level metrics, not accuracy alone.

Transition: Summarise case study findings.

2.11 Slide 11 — Case study conclusion

main.pdf slide: 11 -- Community Hospital

Suggested: 1–2 min

ON SCREEN

Findings block: NLP to SATS before vitals; pathway on every row; 500 Green diversion lever. Plain: next transshipment LP.

SAY THIS

Three takeaways from the case study. First, NLP maps Ghanaian free text to SATS colour **before vitals**—Stewart and Gligorijević on text triage, confirmed in our cohort. Second, every case includes destination and $T_{\max}$ —pathway continuity. Third, five hundred Green diversions are an operational lever on crowding per Yilmaz. Next we model ED flow as a **transshipment network**.

Transition: Introduce the LP framing.

2.12 Slide 12 — ED overcrowding as a network problem

main.pdf slide: 12 -- Transshipment

Suggested: 1–2 min

ON SCREEN

Problem statement; transshipment LP $\min Z = \sum c_{ij}x_{ij}$; supply/balance/demand; plain block on flow metaphor.

THE IDEA

Think of patients as **flow** and hospital zones as **nodes**. A transshipment LP minimises total routing cost: expected wait plus penalty for wrong-queue assignment. The chatbot is a cheap transshipment node that diverts Green patients before they hit majors.

SAY THIS

Overcrowding is not only a queue length statistic—Yilmaz and Mitchell—it is a **routing** problem. When non-urgent patients enter the acute queue, everyone waits longer. We model this as a transshipment linear program: minimise total cost Z over arcs x_{ij} , subject to supply at arrival streams, balance at intermediate nodes, and demand at resuscitation, majors, and OPD. In our Community Hospital instance, cost combines wait and wrong-queue penalty.

Transition: Walk through the network diagram.

2.13 Slide 13 — Hospital transshipment network

main.pdf slide: 13 -- Transshipment

Suggested: 2 min

ON SCREEN

TikZ diagram: three layers (Sources, Transshipment, Destinations). Arrivals (50, 150, 300, 500) → chatbot + desk → resus, majors, OPD.

SAY THIS

Top row: four arrival **sources**—Critical fifty, Very urgent one fifty, Urgent three hundred, Non-urgent five hundred—matching our skewed cohort totals. Middle row: **transshipment nodes**— chatbot pre-triage and the physical triage desk. Bottom row: **destinations**— resuscitation fifty, majors four fifty, OPD five hundred. Trace arcs: most non-urgent flow can pass through the chatbot early; urgent three hundreds may still hit the desk. Point to layer labels on the left so the audience reads top-down.

Transition: Show the LP table with numbers.

2.14 Slide 14 — LP structure

main.pdf slide: 14 -- Transshipment

Suggested: 2 min

ON SCREEN

Decision variables x_{ij} ; supply/demand numbers; block table: without chatbot $Z = 7150$, acute load 900; with chatbot $Z = 2700$, load 500; 62% and 44% reductions.

SAY THIS

Decision variables x_{ij} count patients on each arc. Supplies match arrivals; demands match bed targets. **Without** digital pre-triage, total cost is seven thousand one hundred fifty and **nine hundred** patients load the acute zone—Green mis-routes to majors. **With** the chatbot transshipment node, cost drops to two thousand seven hundred—a **sixty-two percent** reduction—and acute load falls to five hundred—**forty-four percent**. These are analytical results from our Community Hospital case study, not a deployed hospital

optimizer yet.

Transition: Show the scenario chart.

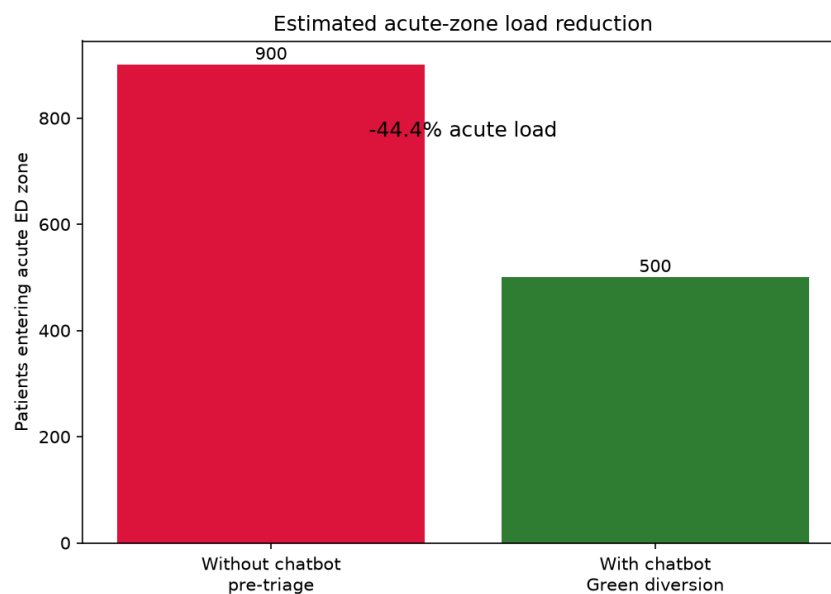
2.15 Slide 15 — Acute-zone load reduction

main.pdf slide: 15 -- Transshipment

Suggested: 1–2 min

ON SCREEN

Chart `wait_time_scenario.png`. Plain: 500 Green patients can mis-route without pre-triage.



SAY THIS

This chart visualises the scenario comparison. Without early Green diversion—Yilmaz’s overcrowding picture—five hundred low-acuity patients can inflate majors queues. With chatbot pre-triage they shift to OPD early in our model. The OR story supports the clinical story: text-based acuity is not only classification; it is a **flow-management** lever.

Transition: Switch to NLP system design.

2.16 Slide 16 — Conversational NLP supports triage

main.pdf slide: 16 -- NLP design

Suggested: 2 min

ON SCREEN

Bullets on chat intake, optional voice/OpenMed; fixed outputs block: $\hat{c}$, $C = f_{\text{SATS}}(\hat{c})$, $P(C)$.

SAY THIS

The chatbot is **conversational**—patients speak in Ghanaian English with local context—Stewart and An et al. on conversational health agents. Optional voice, medical gate, OpenMed tags sit on BioBERT per Lee et al. But output is **not** free-form medical advice—Cheng on triage under uncertainty. Fixed outputs only: predicted acuity one to five; SATS colour via deterministic map; pathway card with destination, T_{max} , actions—Sundström and SATS protocol.

Transition: Contrast with generative LLMs.

2.17 Slide 17 — Versus generative clinical chatbots

main.pdf slide: 17 -- NLP design

Suggested: 2 min

ON SCREEN

Two-column table: Generative LLM vs Curatio—open text vs fixed class; hallucination vs bounded set; audit vs metrics. Plain: nurse remains in the loop.

SAY THIS

Why not ChatGPT? Generative models give open-ended replies—An et al.—with hallucination risk Stewart flags. We output a **bounded set**: class, colour, pathway card—auditable with confidence and per-level metrics in our case study. Instead of “you might have pneumonia,” we say “Orange, acute bed, ten minutes” per SATS. The nurse remains in the loop—Cheng and Rominski—pre-triage and route, not replace clinical judgment.

Transition: Close with summary.

2.18 Slide 18 — Summary

main.pdf slide: 18 -- Conclusion

Suggested: 2 min

ON SCREEN

Four numbered takeaways: gaps; case study numbers; transshipment reductions; NLP design.

SAY THIS

Four closing points. **One:** prior gaps—text before vitals, pathway continuity, overcrowding, constrained NLP, class imbalance—Stewart, Sundström, Mitchell. **Two:** Community Hospital— one thousand skewed cases, ninety-seven point five percent accuracy, pathway on every row. **Three:** transshipment OR—sixty-two percent cost and forty-four percent acute-load reduction with Green diversion. **Four:** NLP design—conversation in, fixed acuity and pathway out, not generative diagnosis—Lee and Cheng.

Transition: Acknowledge limitations honestly.

2.19 Slide 19 — Limitations

main.pdf slide: 19 -- Conclusion

Suggested: 1–2 min

ON SCREEN

Three bullets: synthetic cohort; simulated eval / holdout reference; analytical transshipment only.

SAY THIS

We are transparent on limits. **First**, synthetic cohort—not yet prospective validation at KATH where Rominski validated SATS. **Second**, case-study chatbot eval was simulated when the ML service was offline; BioBERT holdout at ninety-nine point ninety-two percent is the separate reference. **Third**, transshipment is an analytical model—Mitchell’s LMIC triage impact work sets context—not a deployed hospital optimizer today.

Transition: End on references; invite questions.

2.20 Slide 20 — References

main.pdf slide: 20 -- Closing

Suggested: 1 min (do not read every line)

ON SCREEN

Full bibliography from `references.bib` (allow frame breaks).

SAY THIS

Full references are on screen—SATS manual, Rominski, Stewart, Mitchell, Yilmaz, Lee on BioBERT, and our This study twenty twenty-six case-study entry. I will not read every line aloud; the PDF and `references.bib` are available. Thank you—we welcome your questions.

3 Anticipated Q&A

LIKELY QUESTION & ANSWER

Q: Why synthetic data?

A: Real ED data is hard to release; we need a skewed mix—five-fifteen-thirty-twenty-five-twenty-five— to mirror Mitchell on LMIC statistics. BioBERT is validated separately on an eight-thousand holdout at ninety-nine point ninety-two percent.

LIKELY QUESTION & ANSWER

Q: Why not ChatGPT?

A: Bounded acuity class and pathway card; no hallucinated urgency. Stewart's review and our comparison table on slide seventeen—fixed outputs, per-level metrics, nurse in the loop.

LIKELY QUESTION & ANSWER

Q: Why two greens on the chart?

A: Levels four and five are different acuity levels—non-urgent versus routine—but both map to SATS Green per the manual. Distinct bar colours on slide seven; same routing colour.

LIKELY QUESTION & ANSWER

Q: Is TEWS included?

A: Not in this case-study chatbot path—vitals-first gap from slide four. TEWS remains nurse-side; future fusion is documented in `triage-fusion/` at repo root.

4 Rebuild commands

```
cp community-hospital/analysis/figures/*.png community-hospital/slides/figures/  
cd community-hospital/slides  
pdflatex main.tex && bibtex main && pdflatex main.tex && pdflatex main.tex  
cd ../guide  
pdflatex presenter-walkthrough.tex && pdflatex presenter-walkthrough.tex
```